Maximiser votre productivite en rendant votre clavier 2.0 ergonomique

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Mon historique de claviers et dispositions

2009

Clavier classique (105)

2017

Typematrix 2023 (90)

Keyboadio model 1/100 (64)

2026

Corne 42 (42) ?

Azerty

Bépo

Bépo

Ergo-L

Qu’est-ce qui définit un clavier ergonomique ?

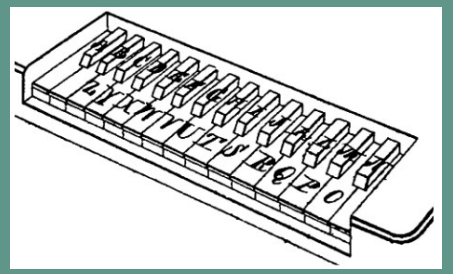
- La disposition clavier (Azerty, Bépo, Ergo-L ...)
- La géométrie physique du clavier
- Disposition des touches autres que les lettres et des actions custom

Les métriques pour une disposition clavier

- Centrer la Heatmap des lettres fréquentes sur les positions confortables
- Répartir la charge des doigts suivant le type de doigt
- Minimiser SFU (Same Finger Usage)
- Minimiser les Ciseaux (CIS) / Extensions (LSB)
- Faciliter Bigrammes fréquents via des roulements intérieurs / extérieurs (RI et RE)
- Minimiser mauvaises redirections (MV)

L’origine du Qwerty :
Machine à écrire Sholes et Glidden

1870

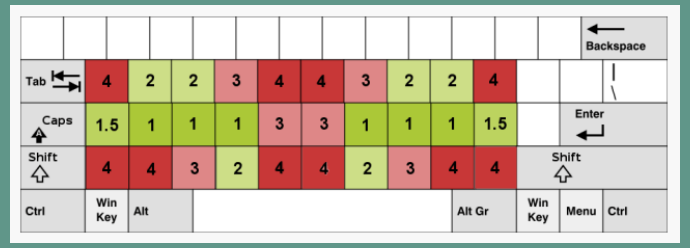


→

1873-1874

Qwerty (1873, en)

SFU↓ 5.74 %
LSB↓ 8.88 %
CIS ↓ 3.53 %
RI ↑ 9.40 %
RO ↑ 8.50 %
MV ↓ 0.45 %



Azerty (18??, fr)

SFU↓ 6.69 %
LSB↓ 8.51 %
CIS ↓ 4.83 %
RI ↑ 8.50 %
RO ↑ 8.90 %
MV ↓ 0.36 %

https://ergol.org/stats/

Dvorak (1936, en)

SFU↓ 2.70 %
LSB↓ 6.57 %
CIS ↓ 0.40 %
RI ↑12.40 %
RO ↑ 5.70 %
MV ↓ 0.52 %

Bépo (2006, fr)

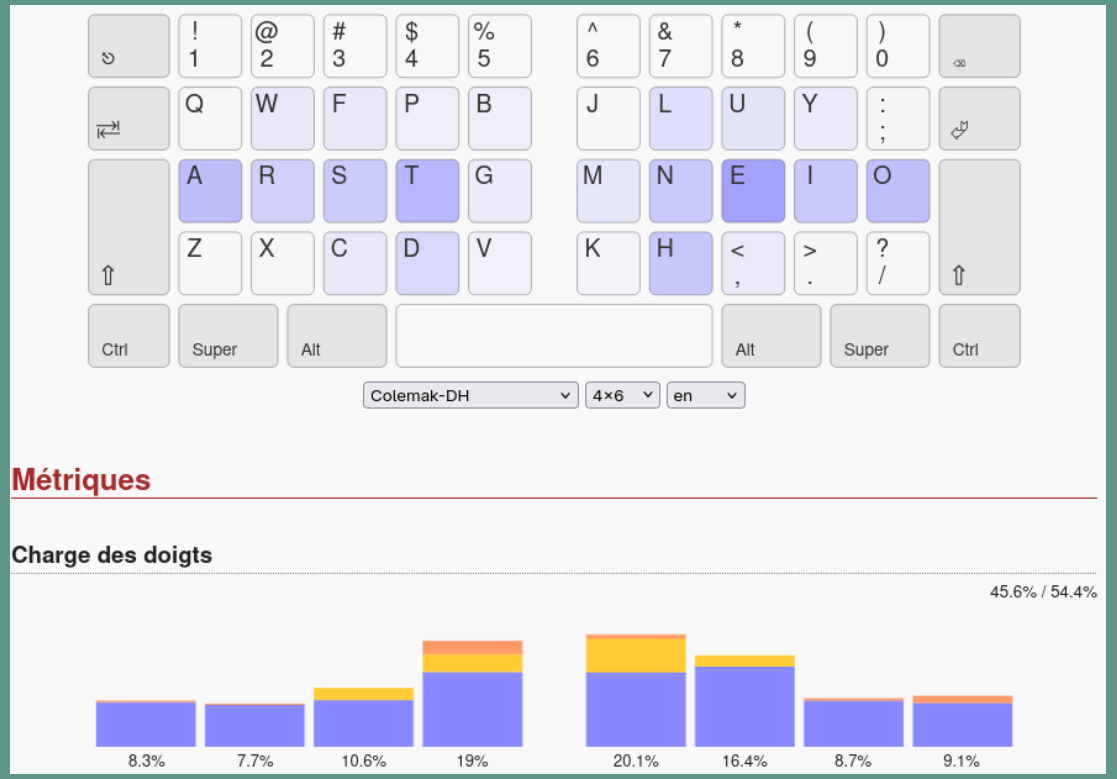
SFU↓ 2.80 %
LSB↓ 3.22 %
CIS ↓ 1.13 %
RI ↑11.60 %
RO ↑ 8.60 %
MV ↓ 0.25 %

- Optimisation texte anglais / français
- Optimisation homerow, haut puis bas
- Alternance des mains

Colemak (2006, en)

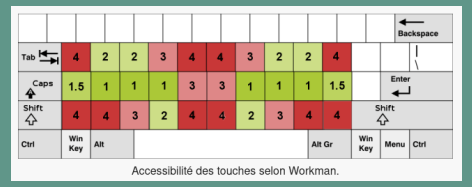
SFU↓ 1.65 %
LSB↓ 9.47 %
CIS ↓ 0.74 %
RI ↑15.90 %
RO ↑10.80 %
MV ↓ 1.12 %

Colemak DH (2014, en)



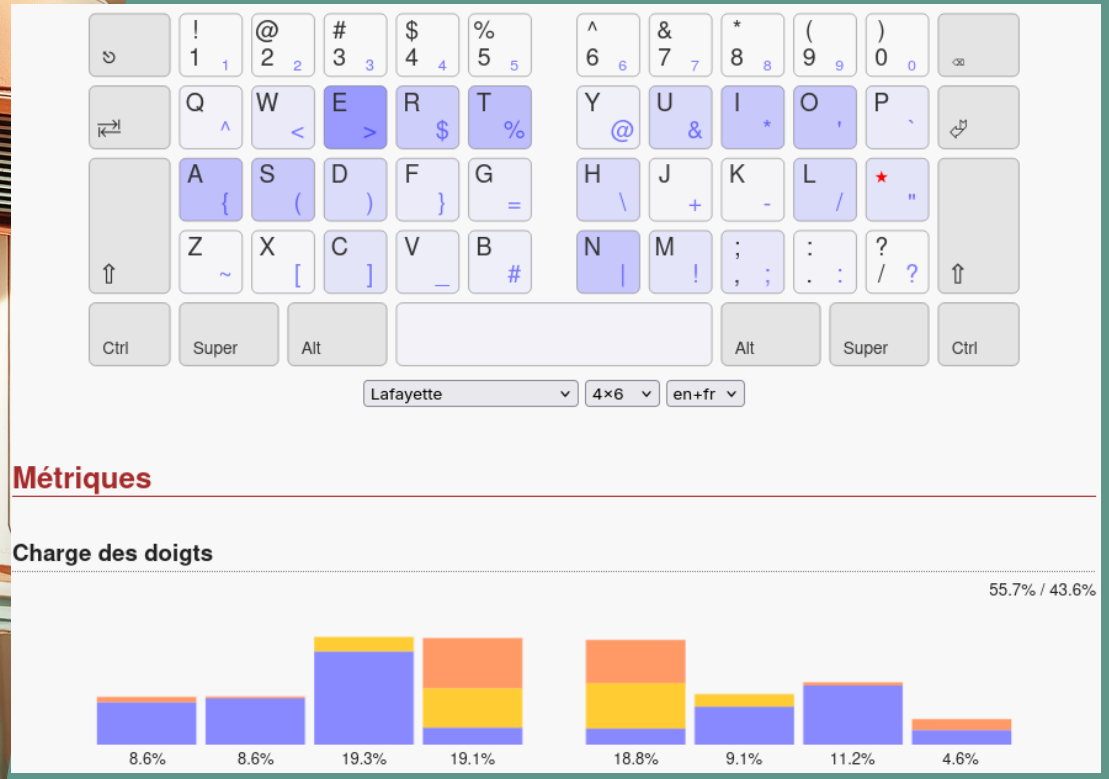
SFU↓ 1.65 %
LSB↓ 5.89 %
CIS ↓ 0.48 %
RI ↑15.10 %
RO ↑15.50 %
MV ↓ 1.12 %

- Roulements des doigts plus qu'alternance
- Conservation des raccourcis à gauche
- DH optimisation Layout Workman



Accessibilité des touches selon Workman.

Qwerty-Lafayette (2010, en+fr) Bépolar (2024, fr)

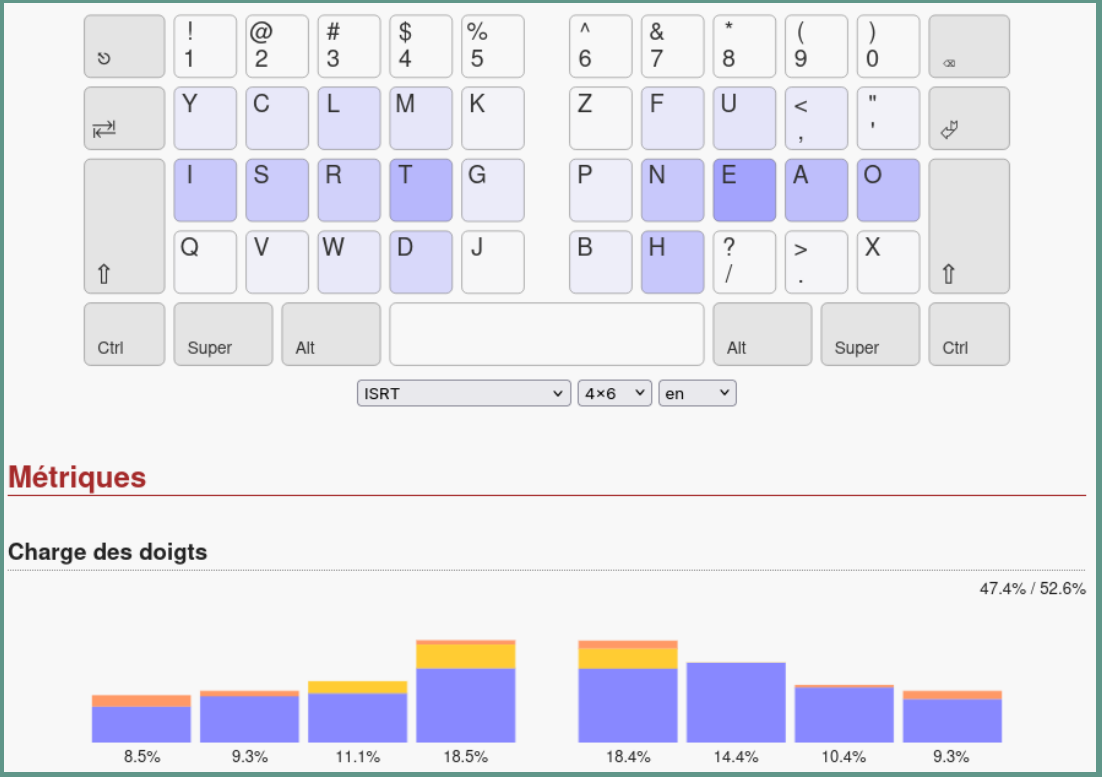


SFU↓ 6.03 %
LSB↓ 7.57 %
CIS ↓ 3.60 %
RI ↑ 9.50 %
RO ↑10.00 %
MV ↓ 0.27 %

- Qwerty/Azerty mais avec une touche morte pour les variantes accentuées
- 1DFH : 1u distance from home
- Symbol programmation via AltGr

SFU↓ 3.64 %
LSB↓ 3.60 %
CIS ↓ 1.14 %
RI ↑13.00 %
RO ↑10.00 %
MV ↓ 0.46 %

MTGAP (2018, anglais), ISRT (2021, anglais)



SFU↓ 1.09 %
LSB↓ 2.47 %
CIS ↓ 1.02 %
RI ↑17.00 %
RO ↑10.00 %
MV ↓ 0.90 %

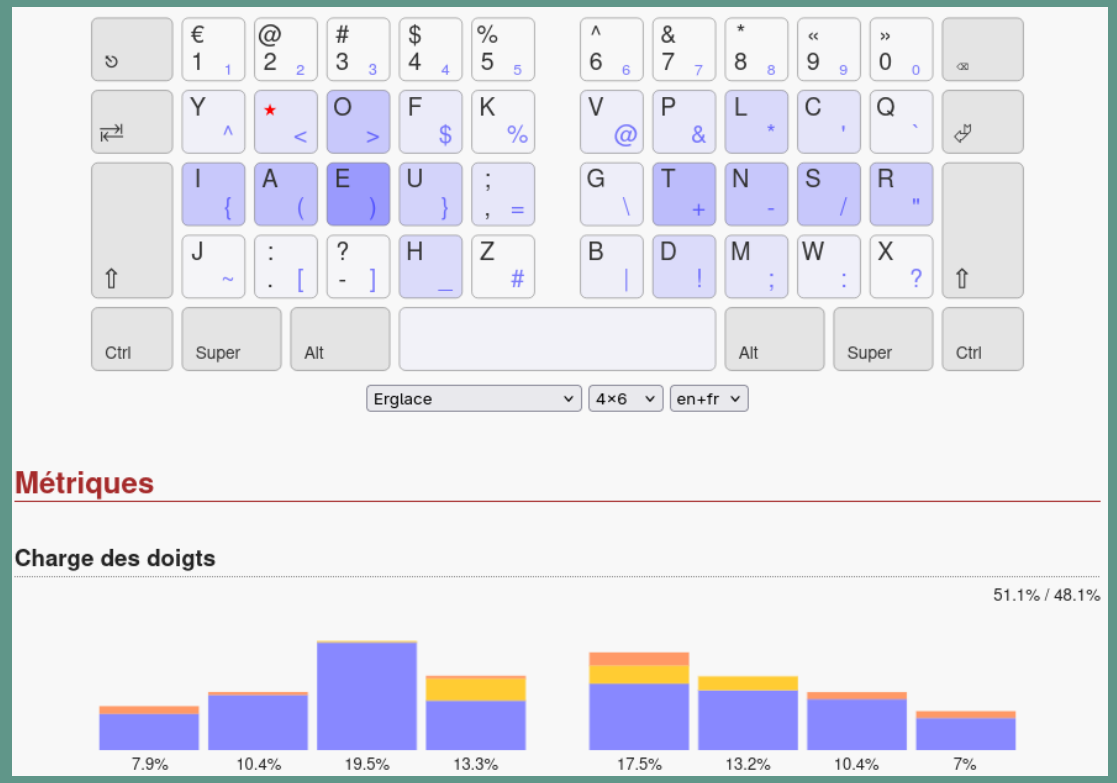
- Plus de raccourcis à gauche
- Roulements des doigts plus qu'alternance

SFU↓ 0.97 %
LSB↓ 4.13 %
CIS ↓ 1.49 %
RI ↑14.60 %
RO ↑16.40 %
MV ↓ 0.64 %

Ergo-L (2024, en+fr)

SFU↓ 1.14 %
LSB↓ 5.73 %
CIS ↓ 1.19 %
RI ↑18.30 %
RO ↑13.90 %
MV ↓ 0.42 %

Erglace (2026 ?, en+fr)



SFU↓ 1.00 %
LSB↓ 3.19 %
CIS ↓ 1.74 %
RI ↑11.40 %
RO ↑11.80 %
MV ↓ 0.20 %

La géométrie physique du clavier

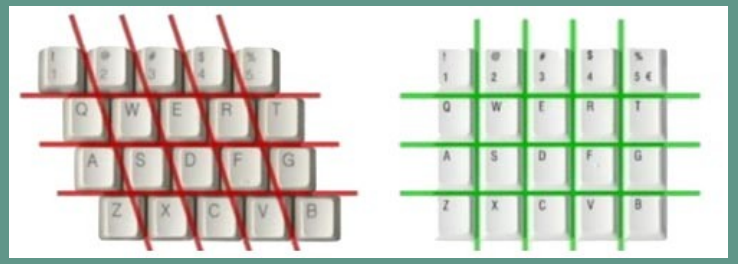
- Ressemblances
  - Disposition Qwerty
  - Alignement décalé en colonnes
  - Rangées sur des plans différents

1874

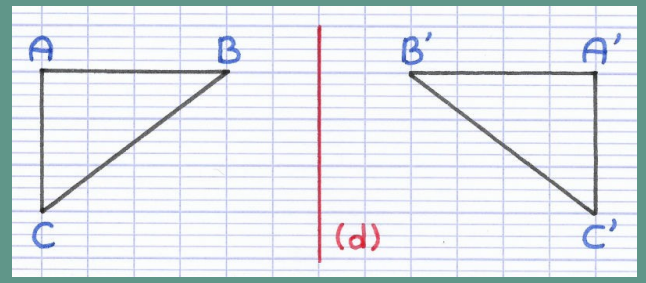
- Différences
  - Modifieurs (ctrl, alt, super)
  - Plus de touches
    - fn, numpad, directions ...

2026

La géométrie physique du clavier



Alignement vertical :
Orthogonal / Column Stagger / Radial



Symétrique

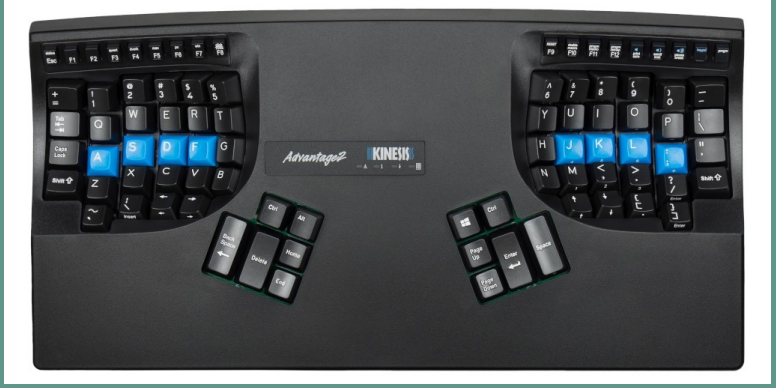
Plat / Cuvette

Accessible / Compact

Clavier full-size (5-6 rangées)

ZSA ErgoDox DZ (2015, 76 keys)

ZSA Moonlander (2020, 72 keys)



Kinesis Avantage 1 / 2 (2002, 88 keys)

MoErgo Glove80 (2025, 88 keys)

Clavier 4x6 (4 rangées)

Keyboardio Model 001 / 100 (2017, 64 keys)

Aurora Lily 58 (2018, 58 keys)

Keyboardio Preonic (2018, 60 keys)

Dygma Defy (2023, 70 keys)

Clavier 3x6 (3 rangées)

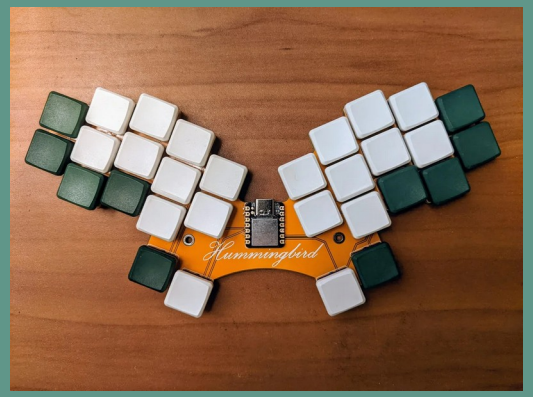
Planck (2016, 48 keys)

Halcyon Corne 42 (2024, 42 keys)

Dispostion 1DFH et home row modifieurs nécessaires

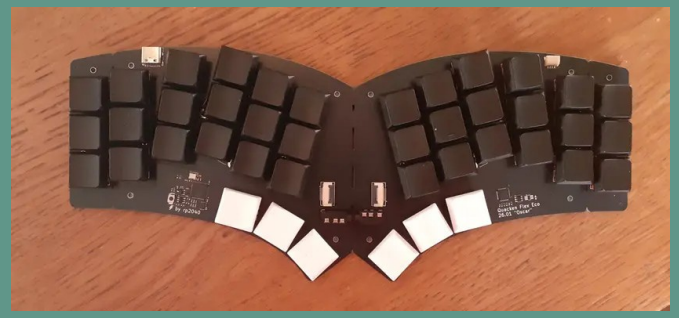
Clavier 3x5 (3 rangées)

Halcyon Ferris 34 (2021, 34 keys)



Humming Bird (2021, 30 keys)

Keyboardio Atreus (2014, 42 keys)



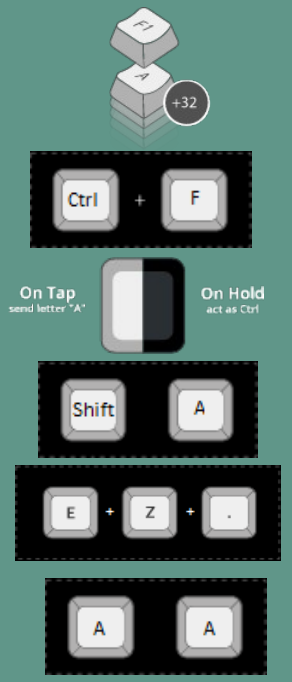
Quacken (2026 ?, 30 → 42 keys)

Dispostion 1DFH et home row modifieurs nécessaires

Configuration / Customisation des touches annexes

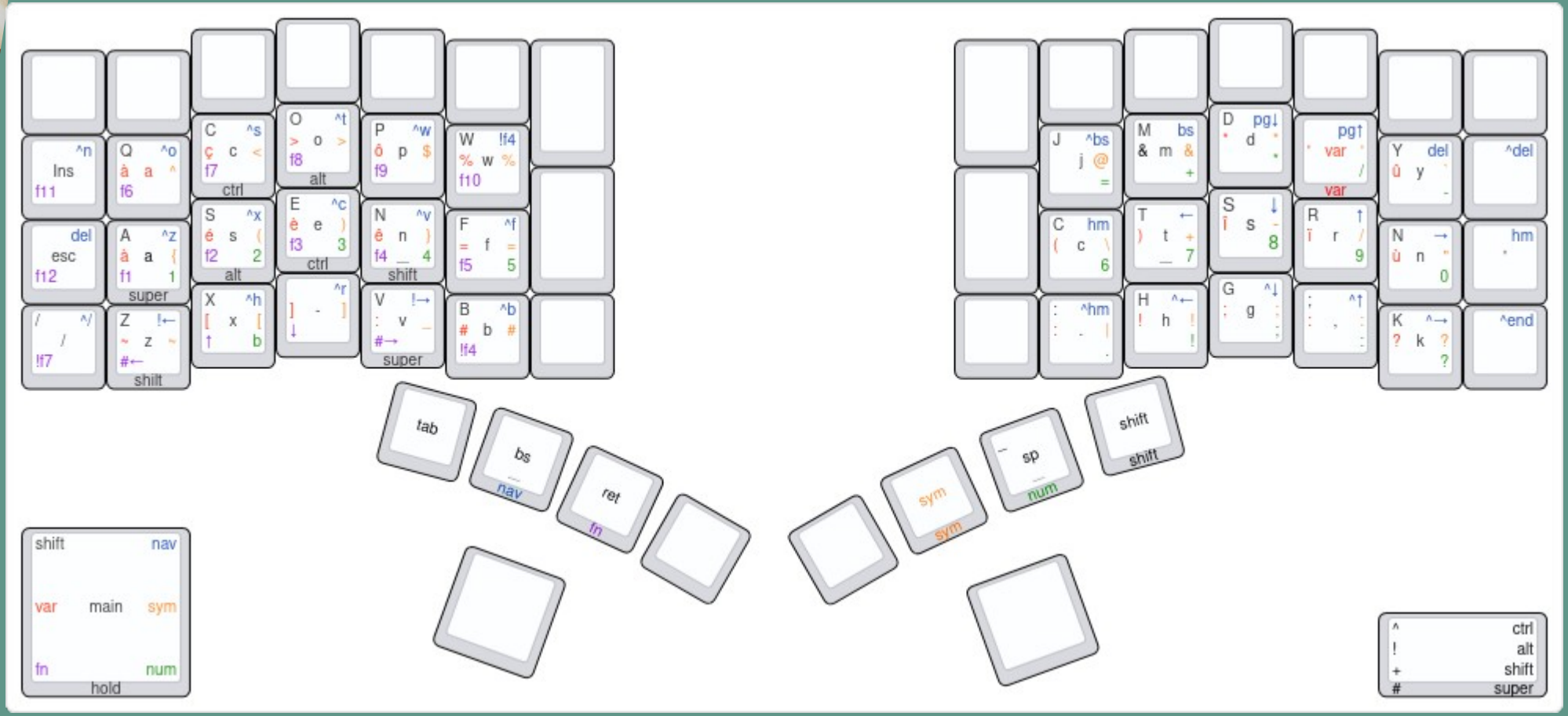
- Hardware (Powered Keyboard Firmware):
  - QMK (2017) / ZMK, Vial (UI)
  - Kaleidoscope (2017), Chrysalis (UI)
- Software :
  - Kanata (2022)

Features de Customistion de clavier

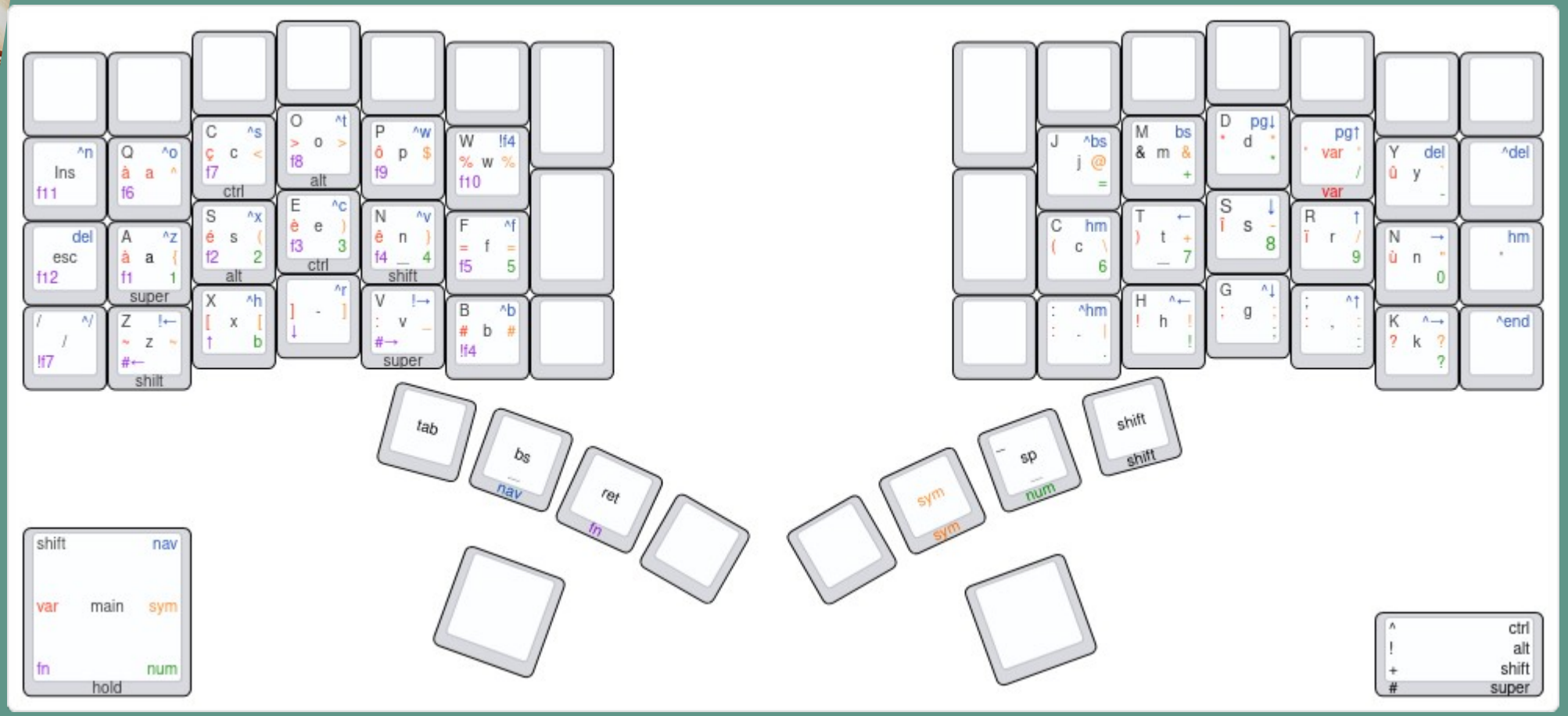


- Layers
- Macro
- Qukeys / Mod Tap
- One Shot
- Combos
- Tap Dance / Repeat Key

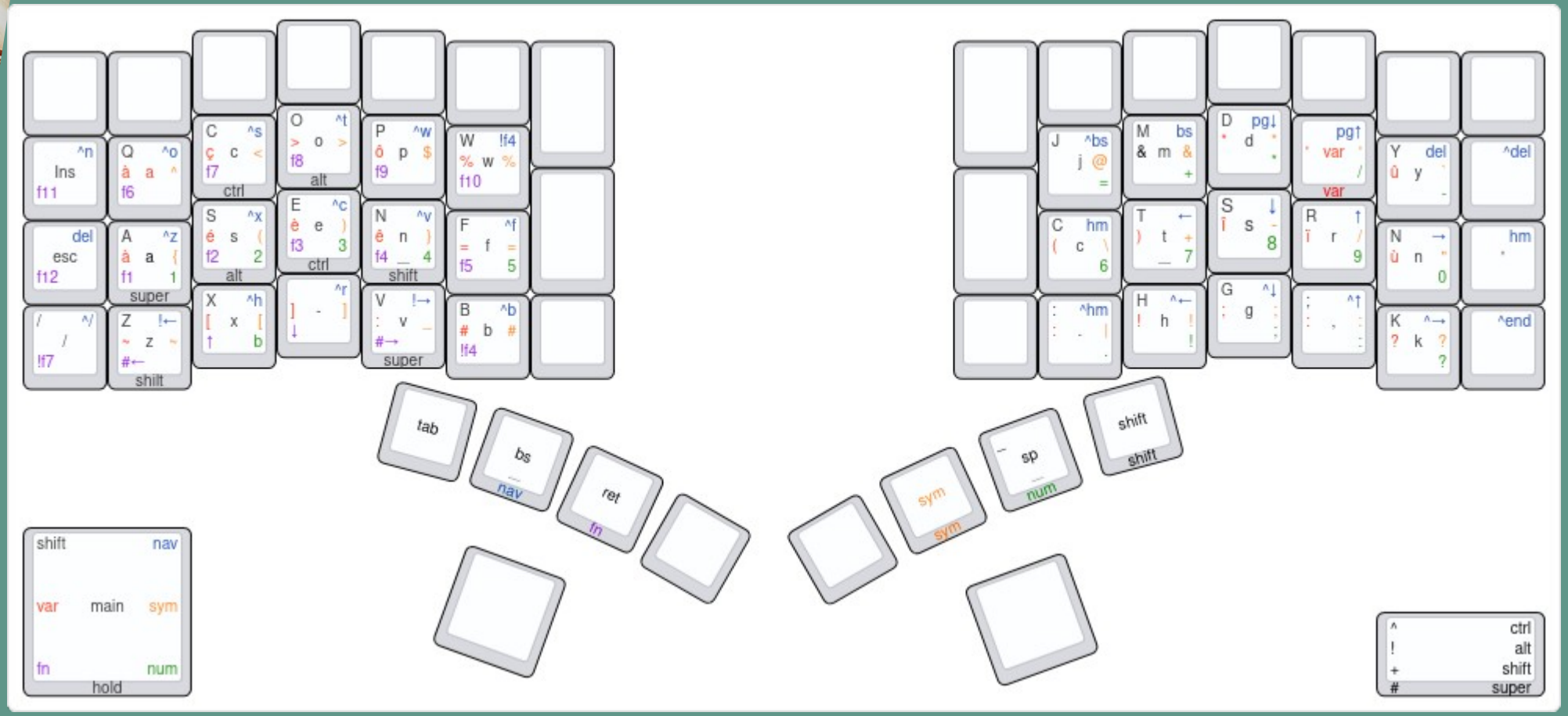
Layer principal



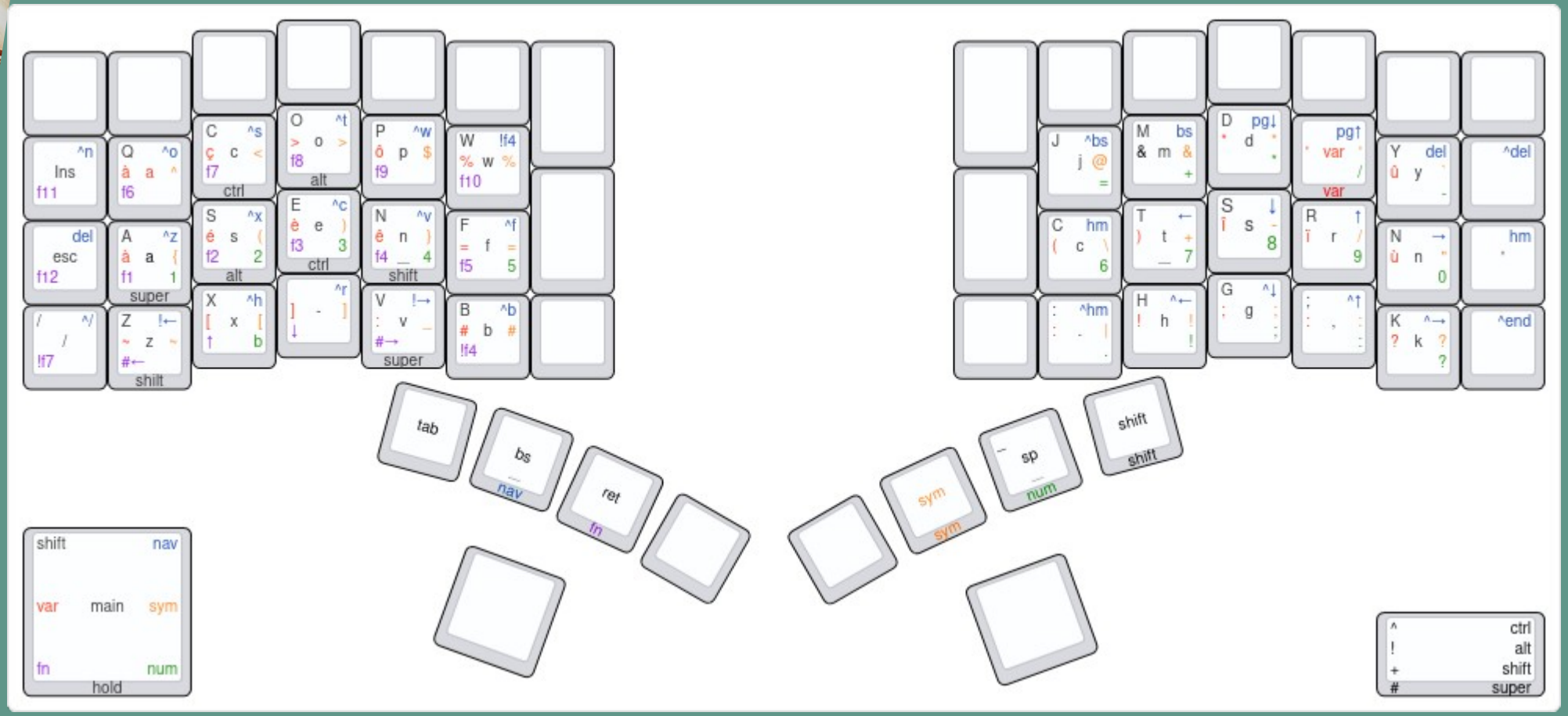
Layer Nav (pouce gauche maintenu)



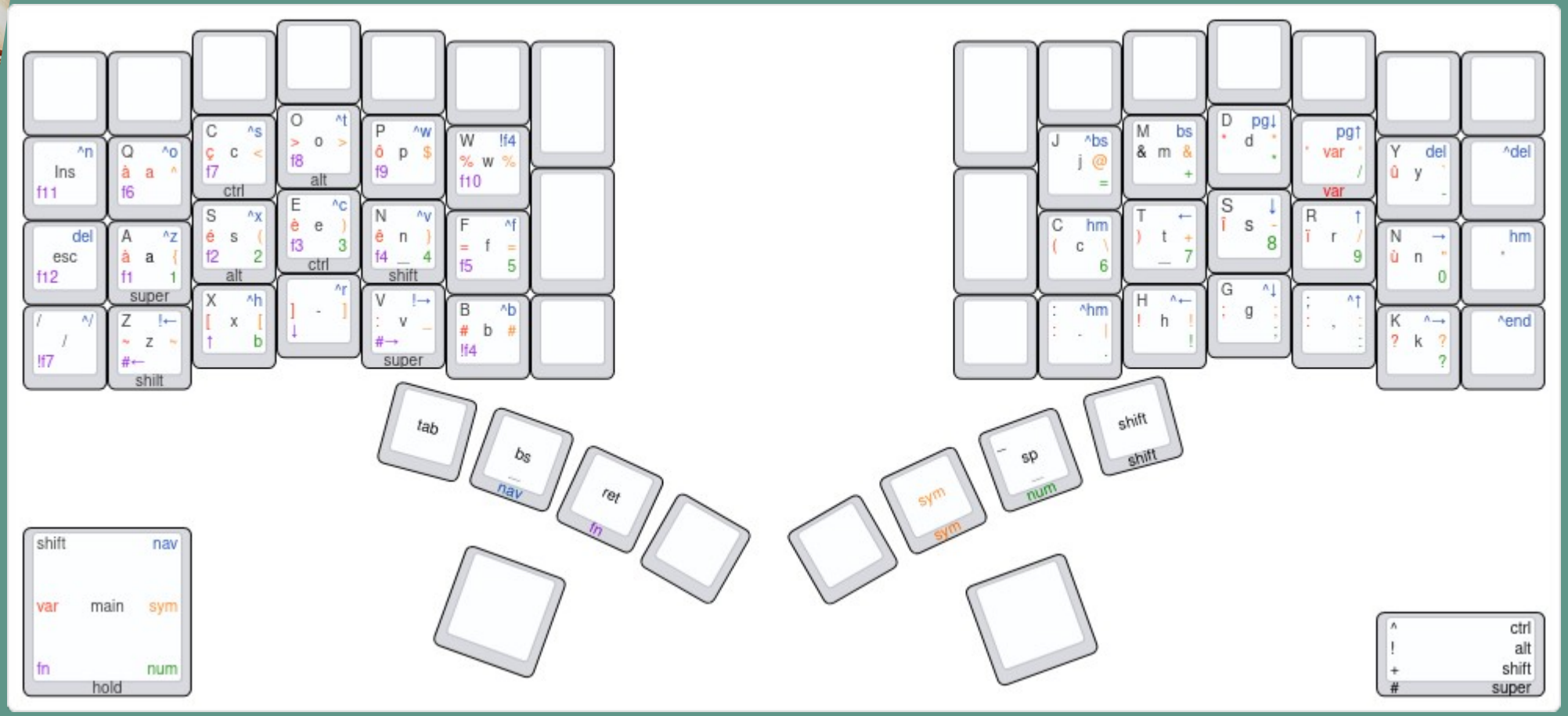
Layer Numrow (pouce droit maintenu)



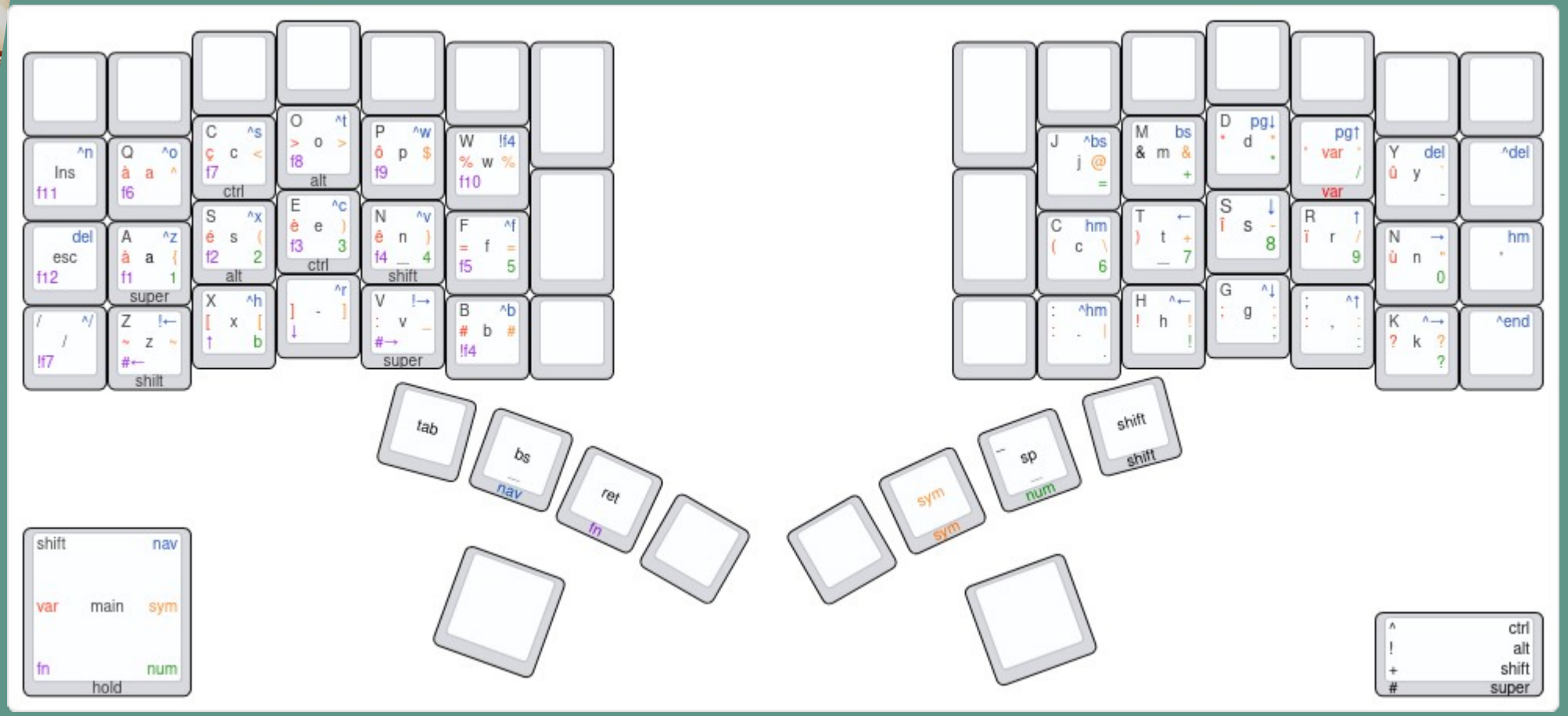
Layer Fn (pouce gauche décalé à droite maintenu)



Layer Symbol (pouce droit décalé à gauche)



Layer Var (Lettre variante, touche morte Ergo-L)



Configuration Kaleidoscope

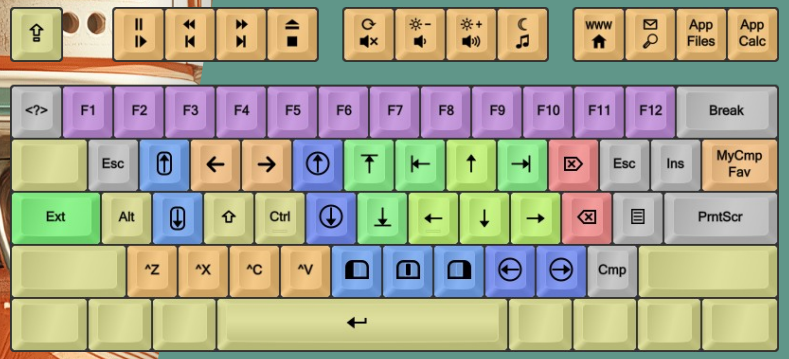
```
[ERGOL_ON_AZERTY_LINUX] = KEYMAP_STACKED
(M(SET_OS_LINUX),                M(SET_OS_MACOS),                M(PRINT_OS),                ___,                        ___,                        ___,                        ___,
 Key_Insert,                     Key_A,                          Key_C,                      Key_O,                      Key_P,                      Key_Z,                      ___,
 Key_Escape,                     Key_Q,                          Key_S,                      Key_E,                      Key_N,                      Key_F,
 Slash,                          Key_W,                          Key_X,                      Minus_sh_Underscore_linux,  Key_V,                      Key_B,                      ___,
 Key_Tab,                        Backspace_x2_CtrlBackspace,     Key_Enter,                  ___,
 ___,

 Key_CapsLock,                   ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 ___,                            Key_J,                          Key_Semicolon,              Key_D,                      OSL(LETTER_VARIANT_LINUX),  Key_Y,                      Key_PcApplication,
                                 Key_L,                          Key_R,                      Key_T,                      Key_I,                      Key_U,                      Apostrophe_sh_QuestionMark,
 ___,                            Period_sh_Colon,                Key_H,                      Key_G,                      Comma_sh_SemiColon,         Key_K,                      Key_PrintScreen,
 ___,                            OSL(SYMBOL_LINUX),              Key_Space,                  OSM(LeftShift),
 ___),

[NAV_LINUX] =  KEYMAP_STACKED
(___,                            ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 LCTRL(Key_N),                   LCTRL(Key_O),                   LCTRL(Key_S),               LCTRL(Key_T),               LCTRL(Key_Z),               LALT(Key_F4),               ___,
 Key_Delete,                     LCTRL(Key_W),                   LCTRL(Key_X),               LCTRL(Key_C),               LCTRL(Key_V),               LCTRL(Key_F),
 LCTRL(Key_KeypadDivide),        LALT(Key_LeftArrow),            LCTRL(Key_H),               LCTRL(Key_R),               LALT(Key_RightArrow),       LCTRL(Key_B),               ___,
 ___,                            ___,                            ___,                        ___,
 ___,

 ___,                            ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 ___,                            LCTRL(Key_Backspace),           Key_Backspace,              Key_PageDown,               Key_PageUp,                 Key_Delete,                 LCTRL(Key_Delete),
                                 Key_Home,                       Key_LeftArrow,              Key_DownArrow,              Key_UpArrow,                Key_RightArrow,             Key_End,
 ___,                            LCTRL(Key_Home),                LCTRL(Key_LeftArrow),       LCTRL(Key_DownArrow),       LCTRL(Key_UpArrow),         LCTRL(Key_RightArrow),      LCTRL(Key_End),
 ___,                            Key_Enter,                      Key_Space,                  ___,
 ___),

[FUNCTION_LINUX] =  KEYMAP_STACKED
(___,                            ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 Key_F11,                        Key_F6,                         Key_F7,                     Key_F8,                     Key_F9,                     Key_F10,                    ___,
 Key_F12,                        Key_F1,                         Key_F2,                     Key_F3,                     Key_F4,                     Key_F5,
 ___,                            LGUI(Key_LeftArrow),            LGUI(Key_UpArrow),          LGUI(Key_DownArrow),        LGUI(Key_RightArrow),       LALT(Key_F4),               ___,
 ___,                            ___,                            ___,                        ___,
 ___,

 ___,                            ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 ___,                            ___,                            ___,                        ___,                        ___,                        ___,                        ___,
                                 ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 ___,                            ___,                            ___,                        ___,                        ___,                        ___,                        ___,
 ___,                            Key_Enter,                      ___,                        ___,
 ___),
```

Ex de cutomisation : Arsenik, Extend, Miryoku, Neo

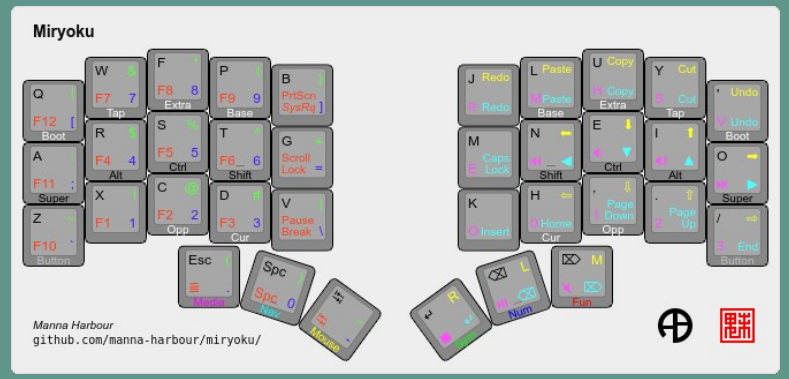
Arsenik

Arsenik Vim variante Layer

Arsenik Num Row Layer



Extend



Miryoku

Neo

Configuration Kanata Arsenik

```
;; Vim-Navigation layer:
;;  - right: Vim-like arrows on HJKL, home/end page up/down, mouse scroll
;;  - left: one-hand shortcuts (Cmd/Ctrl-WASZXCV), Tab/Shift-Tab, prev/next
;;  - top: Super-num (i3/sway) or Alt-num (browser), zoom in/out

;; The `lrld` action stands for "live reload". This will re-parse everything
;; except for linux-dev, i.e. you cannot live reload and switch keyboard devices.

(deflayer navigation
  M-1  M-2  M-3  M-4  M-5  lrld M-6  M-7  M-8  M-9  M-0
  @pad @cls bck  fwd  XX        home pgdn pgup end  @run
  @all @sav S-tab tab XX        lft  down up   rght @fun
  @ndo @cut @cpy @pst XX    _   @mwl @mwd @mwu @mwr XX
            del             _             esc
)

;; NumPad
(deflayer numpad
  _    _    _    _    _     _   _    _    _    _    _
  XX   home up   end  pgup      @/   @7   @8   @9   XX
  XX   lft  down rght pgdn      @-   @4   @5   @6   @0
  XX   XX   XX   XX   XX    _   @,   @1   @2   @3   @.
            @std           @nbs           @std
)

;; function keys
(deflayer funpad
  XX   XX   XX   XX   XX   XX   XX   XX   XX   XX   XX
  f1   f2   f3   f4   XX        XX   XX   XX   XX   XX
  f5   f6   f7   f8   XX        XX   lalt lctl lmet _
  f9   f10  f11  f12  XX   XX   XX   XX   XX   XX   XX
            _               _             _
)
```

Call for Action : Tester une configuration

- Tester une cutomisation avec votre clavier actuel
  - Kanata

et Arsenik

Call for Action : Changer de Clavier ?

- Chercher des claviers suivant vos expérimentations, https://ergol.org/claviers/

Keyboardio

Keyboardio Model 100

splitkb.com

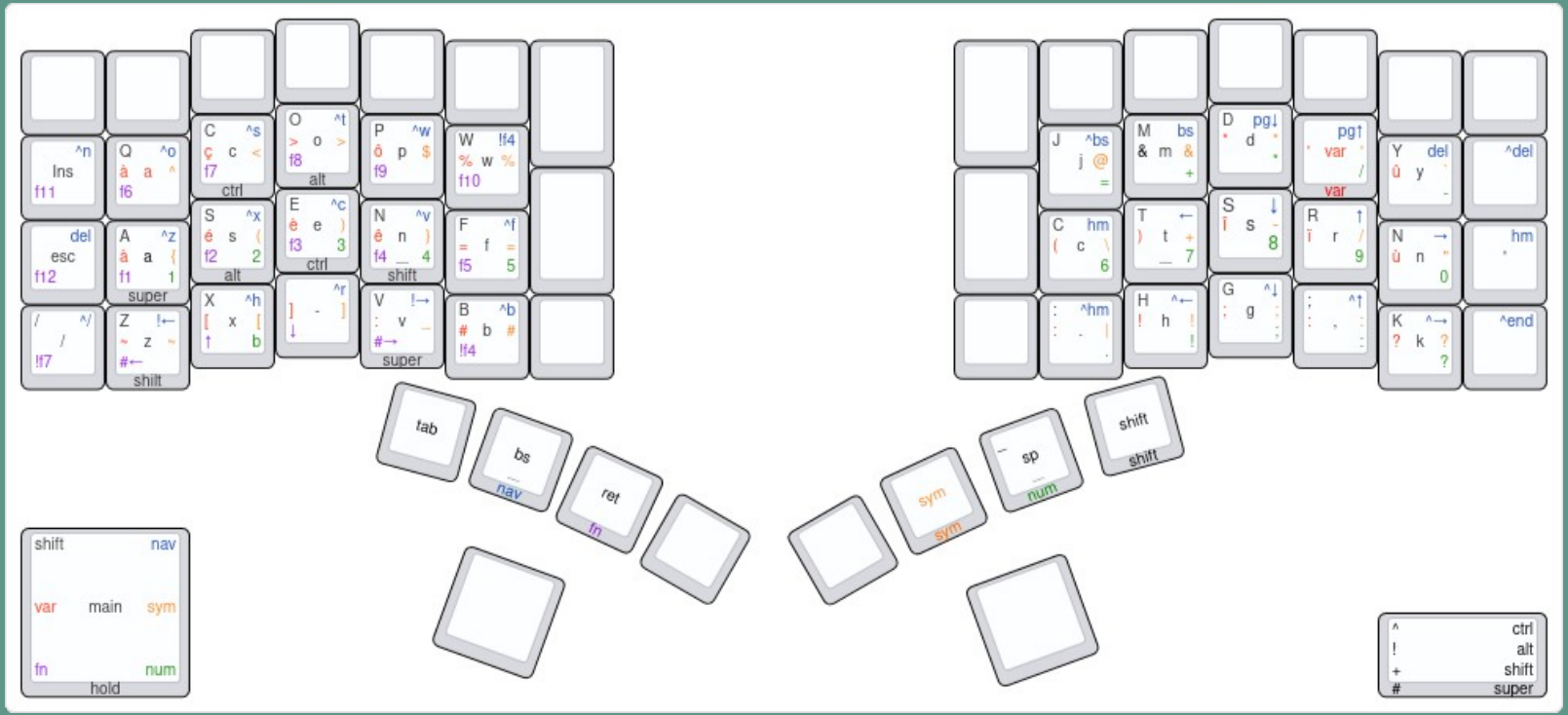
Aurora Lily 58

Halcyon Corne 42

Call for Action : Changer de disposition ?

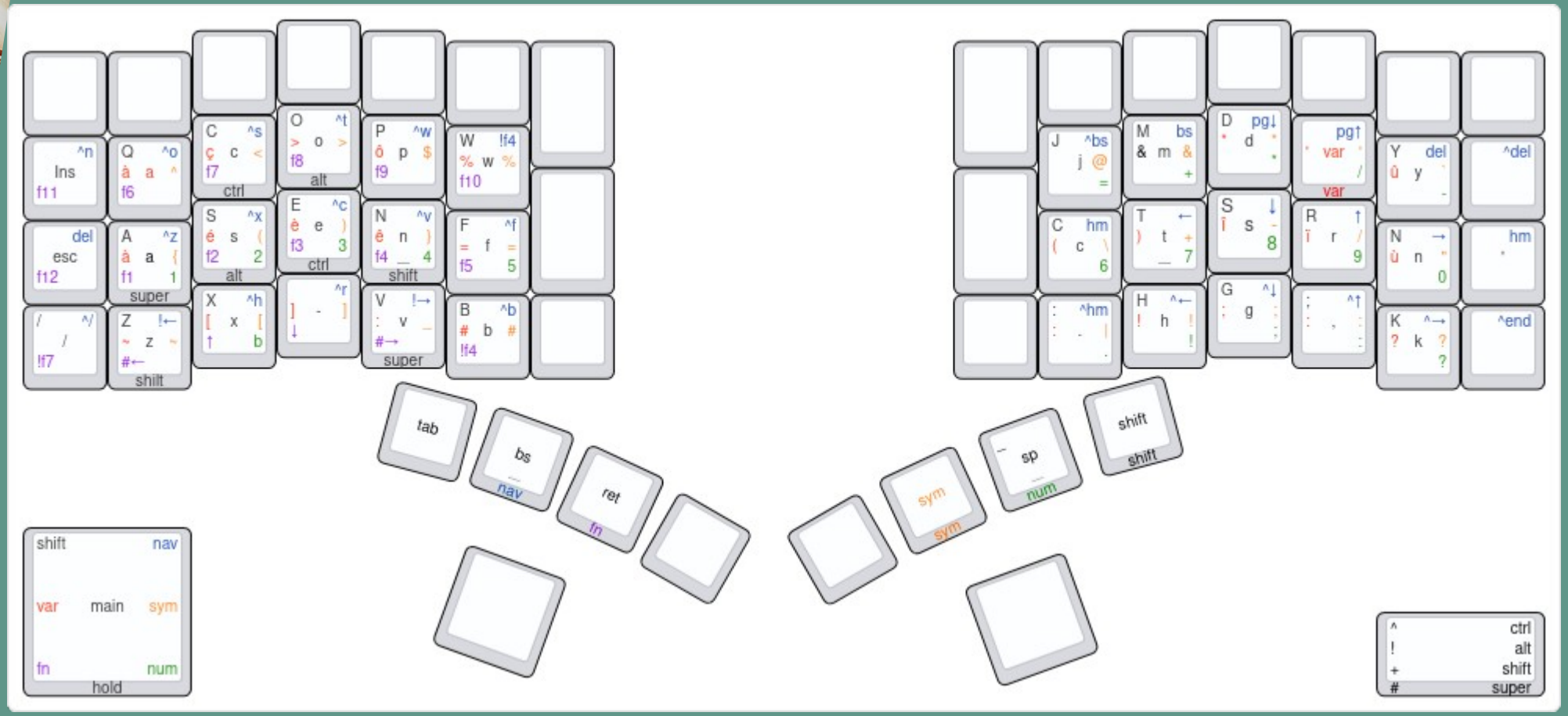
- Apprenez à taper à 10 doigts
- Changer de disposition
  - Qwerty-Lafayette
  - Ergo-L (https://ergol.org/)
- Via
  - Driver OS (droit admin sous windows)
  - Kalamine (AutoHotKey) ou Kanata (droit admin sous windows)
  - Customisation touche clavier (QMK …)
    - Configuration par OS

Questions et liens



alexandrenavarro
alexandre-navarro-tech
alexandrejnavarro.bsky.social
~~alex_j_navarro~~

Annexe : Ma configuration



Annexe : Statistiques